

9.1 Unit description

Introduction

Students are expected to develop experimental skills, and a knowledge and understanding of experimental techniques, by carrying out a range of practical experiments and investigations while they study Units 1 and 2.

This unit will assess students' knowledge and understanding of experimental procedures and techniques that were developed when they conducted these experiments.

Development of practical skills, knowledge and understanding

Students should do a variety of practical work during the IAS course to develop their practical skills. This should help them to gain an understanding and knowledge of the practical techniques that are used in experimental work.

Centres should provide opportunities for students to plan experiments, implement their plans, collect data, analyse their data and draw conclusions in order to prepare them for the assessment of this unit.

Experiments should cover a range of different topic areas and require the use of a variety of practical techniques. The specification for Units 1 and 2 contain suggestions for practical work, although these suggestions do not constitute an exhaustive list. This should help students to gain an understanding and knowledge of the practical techniques that are used in experimental work.

How Science Works

Students should be given the opportunity to develop their practical skills for *How Science Works*, numbers 2–6, as detailed in *Appendix 3*, by completing a range of different experiments that require a variety of different practical techniques.

Students should produce laboratory reports on their experimental work using appropriate scientific, technical and mathematical language, conventions and symbols in order to meet the requirements of *How Science Works*, number 8.